

ANAS MOHAMED OSAMA

NLP & Computer Vision Engineer

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Experience

Digital Egypt Pioneers Initiative (DEPI)

AI Engineer (Microsoft Track)

May 2025 – Nov 2025

Hybrid, Cairo, Egypt

- Gained hands-on experience through 5+ AI/ML projects spanning NLP, Computer Vision, and Predictive Modeling.
- Managed data preparation, model development, evaluation, and deployment workflows on datasets containing 10,000+ records using Microsoft Azure.
- Engineered 10+ AI projects across NLP, Computer Vision, and Predictive Modeling.
- Applied machine learning and deep learning techniques to solve real-world business problems.

Agricultural Bank of Egypt

Summer Intern

Jul 2025 – Aug 2025

Damanhour, Egypt

- Analyzed with structured financial data and banking operations.
- Processed over 500 high-value client checks weekly, ensuring 99.8% accuracy and expediting deposit timelines by two full business days for improved cash flow.
- Managed email communications and assisted in delivering client services.
- Improved data-entry accuracy and gained experience with banking workflows.

IEEE Delta

AI Intern

Oct 2024 – Jan 2025

Remote

- Created and optimized machine learning models achieving up to 95% accuracy on real-world datasets.
- Applied statistical analysis and predictive modeling techniques to financial datasets containing 50,000+ records.

Menofia University

Programming Instructor (Peer Teaching)

2024 – Present

Remote

- Delivered peer-led programming and data science sessions for university students.
- Explained programming fundamentals, statistics, and introductory machine learning concepts.
- Mentored classmates through practical coding exercises and problem-solving sessions.

Projects

DEPI Final Project: Land Type Classification using Sentinel-2 | GitHub

Python, Remote Sensing, Deep Learning, Computer Vision

- Designed and trained a deep learning-based land cover classification system using EuroSAT Sentinel-2 satellite imagery across 10 land-cover classes.
- Performed image preprocessing and feature engineering on 27,000+ satellite image samples for large-scale land cover classification.
- Achieved 92% classification accuracy and 89% F1-score using deep learning and computer vision techniques for terrain categorization.
- Engineered a scalable pipeline for satellite image analysis and environmental monitoring applications.

AI-Vision-Pipeline | GitHub

Python, Computer Vision, Deep Learning

- Architected an end-to-end AI vision pipeline integrating image understanding, story generation, style transformation, and multilingual translation.
- Integrated Azure Computer Vision API, Azure Translator API, and Hugging Face Transformers into a unified automated workflow.
- Orchestrated image analysis, narrative generation, style adaptation, and translation through a modular multi-stage processing pipeline.
- Delivered an interactive Gradio-based web application enabling real-time image-to-story generation and multilingual content delivery.

System Control Using Hand Gesture Detection | GitHub

Python, OpenCV, MediaPipe

- Developed a real-time hand gesture recognition system supporting 10 gesture-based commands for touchless computer interaction.
- Implemented 21-point hand landmark detection and real-time gesture tracking using MediaPipe and OpenCV.
- Automated system operations including cursor navigation, media control, and application interaction through gesture-based commands.

AI News Intelligence System | GitHub

Python, NLP, Machine Learning

- Developed an NLP-based news intelligence system trained on 200,000+ news articles across 15 categories for classification and content analysis.
- Addressed severe class imbalance through data preprocessing and balancing strategies to improve model generalization across all categories.
- Built an ensemble architecture combining DistilBERT and RoBERTa, achieving an F1-score of 84% while preserving fine-grained multi-class categorization.
- Designed a multi-stage news verification pipeline utilizing LLM-based content validation and fake-news detection for credibility assessment.

Education

Menoufia University

Bachelor of Artificial Intelligence

Expected Jul 2027

Data Science Specialization

Leadership Community Involvement

EduGate Student Activity

Vice Head

2023 – 2025

Menoufia University

- Co-led a student educational organization dedicated to academic support and knowledge sharing.
- Managed and coordinated a team of 20+ members responsible for creating study materials, course summaries, and educational content.
- Organized academic sessions, workshops, and mentoring activities to support student learning and engagement.
- Collaborated with team leaders to improve content quality, team productivity, and student outreach initiatives.

Digital Egypt Pioneers Initiative (DEPI)

Team Leader

2025

Leadership Program

- Led and coordinated a student team throughout the DEPI Leadership Program.
- Managed task allocation, team communication, and project coordination activities.
- Recognized as a Top Team Leader and awarded a Leadership Certificate for outstanding performance.

NEX Student Activity

Guest Speaker – Git & GitHub Workshop

2026

Menoufia University

- Delivered a technical workshop on Git and GitHub version control workflows for university students.
- Guided participants through practical GitHub projects and team collaboration exercises.

Certifications

- DEPI AI & Machine Learning Engineer Program (Microsoft Track)
- DEPI Leadership Certificate – Team Leader Recognition
- Agricultural Bank of Egypt – Summer Internship Certificate
- IEEE Delta – AI Internship Certificate

Skills

- **Programming Languages:** Python, C++, SQL, Prolog
- **Frameworks & Libraries:** TensorFlow, PyTorch, Keras, Scikit-learn, NumPy, Pandas, OpenCV, NLTK, Hugging Face Transformers, YOLO
- **Tools & Technologies:** Git, Streamlit, Flask, FastAPI, Linux (Ubuntu), Microsoft Azure
- **Soft Skills:** Leadership, Teamwork, Communication, Research Skills, Presentation Skills, Project Coordination